

THE ZETA FUNCTION OF AN ELLIPTIC CURVE: SOLUTIONS

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Section 1 defines the zeta function of a curve over $\mathbb{F}_q$ as a generating function for its numbers of rational points, and shows that it is also an Euler product over the closed points of the curve. Section 2 computes the zeta function of an elliptic curve E from the action of Frobenius on the Tate module, proves the Riemann hypothesis for it, and deduces Hasse's bound $|\#E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}$. Section 3 shows that the zeta function, equivalently the single number $\#E(\mathbb{F}_q)$, is a complete isogeny invariant.

Reading. [Sil09, §V.2], with [Sil09, §§III.7–III.8] for the Tate module, and [Poo06, §§2.2.1 and 3.4–3.5] for closed points and the Euler product. Three of the problems are exercises from [Sil09, Chapter V]: Problem 2.6 is Exercise 5.3, Problem 2.9 is Exercise 5.13 and Problem 3.2 is Exercise 5.4.

Magma. If you have no local installation, every block below runs in the free [online calculator](#).

Standing notation. Throughout, $q = p^r$ is a power of a prime p , and $\text{Gal}_{\mathbb{F}_q} = \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$. We write $\sigma \in \text{Gal}_{\mathbb{F}_q}$ for the automorphism $t \mapsto t^q$ of $\overline{\mathbb{F}_q}$, which generates $\text{Gal}_{\mathbb{F}_q}$ topologically; its fixed field in $\overline{\mathbb{F}_q}$ is $\mathbb{F}_q$, and that of σ^n is $\mathbb{F}_{q^n}$.

In sections 2 and 3, E is an elliptic curve over $\mathbb{F}_q$, given by a Weierstrass equation with coefficients in $\mathbb{F}_q$. We write $\text{End}(E)$ and $\text{Hom}(E_1, E_2)$ for the endomorphisms and isogenies *defined over* $\mathbb{F}_q$. Then

$$F: E \rightarrow E, \quad (x, y) \mapsto (x^q, y^q),$$

is the q -power Frobenius endomorphism, and

$$a := q + 1 - \#E(\mathbb{F}_q), \quad a_n := q^n + 1 - \#E(\mathbb{F}_{q^n}) \quad (n \geq 1), \quad a_0 := 2,$$

so that $a_1 = a$. For a prime $\ell \neq p$, the Tate module is $T_\ell(E) = \varprojlim_n E[\ell^n](\overline{\mathbb{F}_q})$, a free $\mathbb{Z}_\ell$ -module of rank 2, and $\psi_\ell \in \text{End}(T_\ell(E))$ is the map induced by $\psi \in \text{End}(E)$.

How to use this. Allow about 25 minutes for §1, 35 for §2 and 20 for §3. In Problem 1.2 the heart is parts (2) and (3); parts (5) and (6) can be done at home. Problems 2.6 to 2.8 are the core of §2; Problem 2.9 is short, and Problem 2.10 is a computation to finish at home. In Problem 3.2, part (1) is quick, and the case analysis in parts (4) and (5) is the point of the problem. The Magma parts need no time in the session: do the hand work there and run the code afterwards, as a check on what you wrote. Every block returns in a few seconds.

1. ZETA FUNCTIONS AND CLOSED POINTS

Let C be a smooth projective curve over $\mathbb{F}_q$. A **closed point** P of C is an integral subvariety of dimension 0; its **residue field** $\mathbb{F}_q(P)$ is a finite extension of $\mathbb{F}_q$, and its **degree** is $\deg P := [\mathbb{F}_q(P) : \mathbb{F}_q]$ [Poo06, Definition 2.2.1]. The **zeta function** of C packages the numbers of rational points over all the extensions of $\mathbb{F}_q$ into one power series:

$$Z(C/\mathbb{F}_q; T) := \exp\left(\sum_{n \geq 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n}\right) \in \mathbb{Q}[[T]],$$

so that $\#C(\mathbb{F}_{q^n})$ is the coefficient of T^{n-1} in $\frac{d}{dT} \log Z(C/\mathbb{F}_q; T)$. The exponential looks odd: the more obvious series is $\sum_n \#C(\mathbb{F}_{q^n}) T^n$. Problem 1.2 explains the choice.

Fact 1.1 (Closed points are Galois orbits). Sending a point of $C(\overline{\mathbb{F}}_q)$ to its image is a surjection from $C(\overline{\mathbb{F}}_q)$ onto the set of closed points of C , whose fibers are the $\text{Gal}_{\mathbb{F}_q}$ -orbits; the orbit over a closed point P has $\deg P$ elements. ([Poo06, Remark 2.2.2], where the ground field is perfect throughout, as $\mathbb{F}_q$ is. A point of $C(\overline{\mathbb{F}}_q)$ over P is an $\mathbb{F}_q$ -embedding $\mathbb{F}_q(P) \hookrightarrow \overline{\mathbb{F}}_q$, and there are $\deg P$ of these.)

Problem 1.2 is [Poo06, Proposition 3.4.3 and Exercise 3.7].

Problem 1.2 (The Euler product).

For $d \geq 1$, let N_d be the number of closed points of C of degree d .

- (1) Show that a point of $C(\overline{\mathbb{F}}_q)$ lies in $C(\mathbb{F}_{q^n})$ if and only if σ^n fixes it. Deduce that a $\text{Gal}_{\mathbb{F}_q}$ -orbit of size d is contained in $C(\mathbb{F}_{q^n})$ if $d \mid n$, and is disjoint from it otherwise.
- (2) Show that N_d is finite, and that

$$\#C(\mathbb{F}_{q^n}) = \sum_{d \mid n} d N_d \quad \text{for every } n \geq 1.$$

- (3) Deduce the identity of formal power series

$$Z(C/\mathbb{F}_q; T) = \prod_P (1 - T^{\deg P})^{-1} = \prod_{d \geq 1} (1 - T^d)^{-N_d},$$

the first product running over the closed points P of C . Why is the product well defined? Conclude that $Z(C/\mathbb{F}_q; T) \in 1 + T\mathbb{Z}[[T]]$, which is not obvious from the exponential.

- (4) Invert (2): show that $N_d = \frac{1}{d} \sum_{e \mid d} \mu(d/e) \#C(\mathbb{F}_{q^e})$, where μ is the Möbius function. So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other.
- (5) Take $C = \mathbb{P}^1$. Show that $Z(\mathbb{P}^1/\mathbb{F}_q; T) = 1/((1-T)(1-qT))$. The closed points of $\mathbb{P}^1$ are the point at infinity, of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Deduce that the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$ is $\frac{1}{d} \sum_{e \mid d} \mu(d/e) q^e$. How many are there of degree 2 over $\mathbb{F}_2$, and which are they?
- (6) Put $\zeta_C(s) := Z(C/\mathbb{F}_q; q^{-s})$ and $NP := \# \mathbb{F}_q(P)$. Show that $\zeta_C(s) = \prod_P (1 - NP^{-s})^{-1}$, and compare with the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ of the Riemann zeta function.

Solution.

(1) Embed C in $\mathbb{P}^N$ over $\mathbb{F}_q$ and write a point of $C(\overline{\mathbb{F}}_q)$ as $Q = [x_0 : \cdots : x_N]$ with some $x_i = 1$. Then $\sigma^n Q = [x_0^{q^n} : \cdots : x_N^{q^n}]$, again with i -th coordinate 1, so $\sigma^n Q = Q$ if and only if every x_j lies in $\mathbb{F}_{q^n}$, the fixed field of σ^n , that is, $Q \in C(\mathbb{F}_{q^n})$. The coordinates of Q lie in some finite field $\mathbb{F}_{q^m}$, so the action of $\text{Gal}_{\mathbb{F}_q}$ on the orbit of Q factors through $\text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$, which is generated by the image of σ . Hence the orbit is $\{Q, \sigma Q, \dots, \sigma^{e-1} Q\}$, where σ^e generates the stabilizer of Q in $\langle \sigma \rangle$, and it has e elements; so $e = d$. Thus $\sigma^n Q = Q$ if and only if $d \mid n$, and since $\langle \sigma \rangle$ is abelian, every element of the orbit has the same stabilizer.

(2) By Fact 1.1, a closed point of degree d is an orbit of size d , which by (1) lies in the finite set $C(\mathbb{F}_{q^d})$; so $N_d \leq \#C(\mathbb{F}_{q^d}) < \infty$. The set $C(\mathbb{F}_{q^n})$ is a union of orbits, namely of the orbits of size d for $d \mid n$, by (1). There are N_d of these, each with d elements, so $\#C(\mathbb{F}_{q^n}) = \sum_{d \mid n} d N_d$.

(3) Since N_d is finite, only finitely many factors of the product are $\not\equiv 1 \pmod{T^m}$ for each m , namely those with $\deg P < m$; so the product converges in $\mathbb{Z}[[T]]$, and each factor $(1 - T^d)^{-1} = 1 + T^d + T^{2d} + \dots$ has integer coefficients. Taking logarithms and using $-\log(1-x) = \sum_{k \geq 1} x^k/k$,

$$\log \prod_{d \geq 1} (1 - T^d)^{-N_d} = \sum_{d \geq 1} N_d \sum_{k \geq 1} \frac{T^{dk}}{k} = \sum_{n \geq 1} \left(\sum_{d|n} d N_d \right) \frac{T^n}{n} = \sum_{n \geq 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n},$$

with $n = dk$, so that $1/k = d/n$. Exponentiating gives the claim, and the product has coefficients in $\mathbb{Z}$ and constant term 1.

(4) Put $f(n) = \#C(\mathbb{F}_{q^n})$ and $g(d) = dN_d$. Then $f(n) = \sum_{d|n} g(d)$, and Möbius inversion gives $g(d) = \sum_{e|d} \mu(d/e) f(e)$.

(5) $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1$, so $\log Z(\mathbb{P}^1/\mathbb{F}_q; T) = \sum_n (1 + q^n) T^n/n = -\log(1 - T) - \log(1 - qT)$. By (4) applied to $\mathbb{P}^1$, for $d > 1$ the point at infinity (the only closed point of $\mathbb{P}^1$ outside $\mathbb{A}^1$, of degree 1) does not count, and the extra $+1$ in $\#\mathbb{P}^1(\mathbb{F}_{q^e}) = q^e + 1$ contributes $\frac{1}{d} \sum_{e|d} \mu(d/e) = 0$; so for $d > 1$, and directly for $d = 1$, the number of monic irreducible polynomials of degree d is $\frac{1}{d} \sum_{e|d} \mu(d/e) q^e$. For $q = 2$ and $d = 2$: $\frac{1}{2}(4 - 2) = 1$, the polynomial $t^2 + t + 1$. It is one closed point of degree 2, the orbit $\{\omega, \omega^2\}$ of a primitive cube root of unity in $\mathbb{F}_4$.

(6) Substitute $T = q^{-s}$: $T^{\deg P} = (q^{\deg P})^{-s} = NP^{-s}$, since $\mathbb{F}_q(P)$ has $q^{\deg P}$ elements. The closed points play the role of the primes, and NP that of $p = \#\mathbb{F}_p$.

Remark 1.3 (The Weil conjectures for curves). Let C be a smooth projective curve of genus g over $\mathbb{F}_q$. The Weil conjectures (proved for curves by Weil) say:

- (1) (Rationality) $Z(C/\mathbb{F}_q; T) = P(T)/((1 - T)(1 - qT))$ with $P \in \mathbb{Z}[T]$ of degree $2g$ and $P(0) = 1$;
- (2) (Functional equation) $Z(C/\mathbb{F}_q; 1/(qT)) = q^{1-g} T^{2-2g} Z(C/\mathbb{F}_q; T)$;
- (3) (Riemann hypothesis) $P(T) = \prod_{i=1}^{2g} (1 - \alpha_i T)$ with $|\alpha_i| = \sqrt{q}$ for every i .

For $C = \mathbb{P}^1$, of genus 0, Problem 1.2(5) gives all three with $P = 1$. We prove them for $g = 1$ in section 2.

2. THE ZETA FUNCTION OF AN ELLIPTIC CURVE

What we may use

Let E be an elliptic curve over $\mathbb{F}_q$ and $\ell \neq p$ a prime.

Fact 2.1 (Frobenius is purely inseparable). F is purely inseparable of degree q [Sil09, Proposition II.2.11].

Fact 2.2 ($1 - F$ is separable). $1 - F$ is separable [Sil09, Corollary III.5.5].

Fact 2.3 (Kernels of separable isogenies). A nonzero separable isogeny ψ satisfies $\#\ker \psi = \deg \psi$, where $\ker \psi$ is taken in $E(\overline{\mathbb{F}}_q)$ [Sil09, Theorem III.4.10(c)].

Fact 2.4 (Determinant and trace on $T_\ell(E)$). For $\psi \in \text{End}(E)$, $\det \psi_\ell = \deg \psi$ and $\text{tr} \psi_\ell = 1 + \deg \psi - \deg(1 - \psi)$ [Sil09, Proposition III.8.6].

Fact 2.5 (Endomorphisms act faithfully on $T_\ell(E)$). The map $\psi \mapsto \psi_\ell$ is an injective ring homomorphism $\text{End}(E) \rightarrow \text{End}(T_\ell(E))$ [Sil09, Theorem III.7.4].

Problem 2.6 is [Sil09, Exercise 5.3].

Problem 2.6 (Traces and determinants).

Let A be a square matrix over a field K of characteristic 0. We prove, in two ways, that

$$(2.1) \quad \exp\left(\sum_{n \geq 1} \operatorname{tr}(A^n) \frac{T^n}{n}\right) = \frac{1}{\det(1 - AT)} \quad \text{in } K[[T]].$$

- (1) Let $f, g \in 1 + TK[[T]]$. Show that f and g are units of $K[[T]]$, so that their **logarithmic derivatives** f'/f and g'/g are defined, and that if $f'/f = g'/g$, then $f = g$.^a Where do you use that K has characteristic 0? Give a counterexample in characteristic p .
- (2) Show that both sides of (2.1) lie in $1 + TK[[T]]$, compute their logarithmic derivatives, and conclude with (1).^b
- (3) Give a second proof: triangularize A over an algebraic closure $\bar{K}$ of K , reduce (2.1) to the case of a 1×1 matrix $A = (\lambda)$, and prove that case with (1), applied over $\bar{K}$.

^aHint: compute the derivative of f/g .

^bHint: for $M(T)$ an invertible matrix over $K[[T]]$, $\frac{d}{dT} \det M(T) = \det M(T) \operatorname{tr}(M(T)^{-1} M'(T))$.

Solution.

(1) A power series with constant term 1 is a unit: if $f = 1 - u$ with $u \in TK[[T]]$, then $f^{-1} = \sum_{k \geq 0} u^k$, which converges in $K[[T]]$ because $u^k \in T^k K[[T]]$. Put $h = f/g \in 1 + TK[[T]]$. By the quotient rule,

$$h' = \frac{f'g - fg'}{g^2} = h \left(\frac{f'}{f} - \frac{g'}{g} \right) = 0.$$

Writing $h = \sum_n h_n T^n$, this says $nh_n = 0$ for every $n \geq 1$. Since K has characteristic 0, n is invertible in K , so $h_n = 0$ for $n \geq 1$, and $h = h_0 = 1$. This is the only use of characteristic 0, and it is essential: in characteristic p , $f = 1 + T^p$ and $g = 1$ have $f' = pT^{p-1} = 0 = g'$, so both logarithmic derivatives vanish, but $f \neq g$.

(2) For $F \in TK[[T]]$, $\exp F$ has constant term 1, and the chain rule gives $(\exp F)' = F' \exp F$; so the left side lies in $1 + TK[[T]]$, with logarithmic derivative $\sum_{n \geq 1} \operatorname{tr}(A^n) T^{n-1}$. The right side has constant term $1/\det(1) = 1$. The matrix $M(T) = 1 - AT$ is invertible over $K[[T]]$, with inverse $\sum_{k \geq 0} A^k T^k$, and $M'(T) = -A$; so the logarithmic derivative of $\det(1 - AT)^{-1}$ is

$$-\frac{\frac{d}{dT} \det(1 - AT)}{\det(1 - AT)} = -\operatorname{tr}((1 - AT)^{-1}(-A)) = \operatorname{tr}\left(\sum_{k \geq 0} A^{k+1} T^k\right) = \sum_{n \geq 1} \operatorname{tr}(A^n) T^{n-1}.$$

The two agree, so the two sides are equal by (1).

(3) Over $\bar{K}$, $A = BUB^{-1}$ with U upper triangular, with diagonal entries the eigenvalues $\lambda_1, \dots, \lambda_m$ of A . Then $A^n = BU^n B^{-1}$, and U^n is upper triangular with diagonal entries λ_i^n , so $\operatorname{tr}(A^n) = \sum_i \lambda_i^n$; likewise $1 - AT = B(1 - UT)B^{-1}$, so $\det(1 - AT) = \prod_i (1 - \lambda_i T)$. Since $\exp(F + G) = \exp F \exp G$, both sides of (2.1) are products over i of the corresponding sides for the 1×1 matrix (λ_i) . For $A = (\lambda)$, both $\exp(\sum_n \lambda^n T^n/n)$ and $(1 - \lambda T)^{-1}$ lie in $1 + T\bar{K}[[T]]$, with logarithmic derivatives $\sum_{n \geq 1} \lambda^n T^{n-1}$ and $\lambda/(1 - \lambda T) = \sum_{n \geq 1} \lambda^n T^{n-1}$; so they are equal by (1) over $\bar{K}$, which also has characteristic 0. Both sides of (2.1) have coefficients in K , so the identity in $\bar{K}[[T]]$ is the one we want.

Problem 2.7 is the proof of [Sil09, Theorems V.2.3.1 and V.2.4], arranged around Problem 2.6.

Problem 2.7 (The zeta function of E).

- (1) Show that $\sigma(Q) = F(Q)$ for every $Q \in E(\overline{\mathbb{F}}_q)$. Deduce that the closed points of E of degree d are the F -orbits of size d in $E(\overline{\mathbb{F}}_q)$, and that $E(\mathbb{F}_{q^n}) = \ker(1 - F^n)$.
- (2) Show that F^n is the q^n -power Frobenius endomorphism of E over $\mathbb{F}_{q^n}$, and deduce that

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - F^n) = \det(1 - F_\ell^n) = 1 - \text{tr}(F_\ell^n) + q^n.$$

Where does the argument use that $1 - F^n$ is separable?

- (3) Taking $n = 1$, show that the characteristic polynomial of F_ℓ is $T^2 - aT + q$. In particular it has integer coefficients and does not depend on ℓ .
- (4) Use Problem 2.6 to show that

$$Z(E/\mathbb{F}_q; T) = \frac{\det(1 - F_\ell T)}{(1 - T)(1 - qT)} = \frac{1 - aT + qT^2}{(1 - T)(1 - qT)}.$$

By Problem 1.2, this is also $\prod_P (1 - T^{\deg P})^{-1}$ over the closed points P of E .

- (5) Verify the functional equation $Z(E/\mathbb{F}_q; 1/(qT)) = Z(E/\mathbb{F}_q; T)$ of Remark 1.3.

Solution.

(1) For $Q = (x_0, y_0)$ with $x_0, y_0 \in \overline{\mathbb{F}}_q$, $F(Q) = (x_0^q, y_0^q) = \sigma(Q)$, and both fix O . So the $\text{Gal}_{\overline{\mathbb{F}}_q}$ -orbits are the F -orbits (Problem 1.2(1): the orbit of Q is $\{Q, \sigma Q, \sigma^2 Q, \dots\}$), and Fact 1.1 identifies them with the closed points. By Problem 1.2(1), $Q \in E(\mathbb{F}_{q^n})$ if and only if $F^n(Q) = \sigma^n(Q) = Q$, that is, $Q \in \ker(1 - F^n)$.

(2) $F^n(x, y) = (x^{q^n}, y^{q^n})$, and E is defined over $\mathbb{F}_{q^n}$. So Fact 2.2, applied to E over $\mathbb{F}_{q^n}$, says that $1 - F^n$ is separable; it is nonzero, since its kernel $E(\mathbb{F}_{q^n})$ is finite while $E(\overline{\mathbb{F}}_q)$ is infinite. By Fact 2.3 and (1), $\#E(\mathbb{F}_{q^n}) = \#\ker(1 - F^n) = \deg(1 - F^n)$; this is where separability is used, since for an inseparable isogeny $\#\ker$ is only the separable degree. By Facts 2.4 and 2.5, $\deg(1 - F^n) = \det((1 - F^n)_\ell) = \det(1 - F_\ell^n)$. For a 2×2 matrix M , $\det(1 - M) = 1 - \text{tr } M + \det M$, and $\det F_\ell^n = (\det F_\ell)^n = (\deg F)^n = q^n$ by Fact 2.1.

(3) For a 2×2 matrix the characteristic polynomial is $T^2 - (\text{tr } M)T + \det M$. By (2) with $n = 1$, $\text{tr } F_\ell = 1 + q - \#E(\mathbb{F}_q) = a$, and $\det F_\ell = q$.

(4) By (2), $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \text{tr}(F_\ell^n)$, so, working in $\mathbb{Q}_\ell[[T]]$,

$$Z(E/\mathbb{F}_q; T) = \exp\left(\sum_n \frac{T^n}{n}\right) \exp\left(\sum_n \frac{(qT)^n}{n}\right) \exp\left(-\sum_n \text{tr}(F_\ell^n) \frac{T^n}{n}\right) = \frac{\det(1 - F_\ell T)}{(1 - T)(1 - qT)},$$

using Problem 2.6 over $K = \mathbb{Q}_\ell$ for the last factor. By (3), $\det(1 - F_\ell T) = T^2 \det(T^{-1} - F_\ell) = 1 - aT + qT^2$.

(5) Substituting $T \mapsto 1/(qT)$ and multiplying numerator and denominator by qT^2 :

$$\frac{1 - a/(qT) + 1/(qT^2)}{(1 - 1/(qT))(1 - 1/T)} = \frac{qT^2 - aT + 1}{(qT - 1)(T - 1)} = Z(E/\mathbb{F}_q; T),$$

which is Remark 1.3(2) with $g = 1$.

Problem 2.8 (The Riemann hypothesis and Hasse's bound).

Let $\alpha, \beta \in \mathbb{C}$ be the roots of $T^2 - aT + q$, so that $1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T)$.

- (1) For $r, s \in \mathbb{Z}$ with $s \neq 0$, show that $\det(r/s - F_\ell) = \deg(r - sF)/s^2 \geq 0$. Deduce that $T^2 - aT + q$ is nonnegative on $\mathbb{R}$, and hence that $\beta = \bar{\alpha}$ and $|\alpha| = |\beta| = \sqrt{q}$. This is the

Riemann hypothesis of Remark 1.3. With ζ_E as in Problem 1.2(6), show that every zero of $\zeta_E(s)$ satisfies $\operatorname{Re}(s) = 1/2$, and that $\zeta_E(s) = \zeta_E(1-s)$.

- (2) Show that $a_n = \alpha^n + \beta^n$ for every $n \geq 0$. Deduce Hasse's bound

$$|\#E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q},$$

and more generally $|a_n| \leq 2q^{n/2}$ for every $n \geq 1$.

- (3) Conversely, suppose only that there is a constant c with $|a_n| \leq cq^{n/2}$ for all $n \geq 1$. Show that the power series $\sum_{n \geq 1} a_n T^n = \frac{\alpha T}{1-\alpha T} + \frac{\beta T}{1-\beta T}$ converges for $|T| < q^{-1/2}$, and deduce that $|\alpha| = |\beta| = \sqrt{q}$. So the Riemann hypothesis is equivalent to a Hasse bound over every extension of $\mathbb{F}_q$, even with a worse constant.

Solution.

(1) By Fact 2.5, $(r-sF)_\ell = r-sF_\ell$, so by Fact 2.4 $s^2 \det(r/s - F_\ell) = \det(r-sF_\ell) = \deg(r-sF) \geq 0$. So the real polynomial $T^2 - aT + q = \det(T - F_\ell)$ (Problem 2.7(3)) is ≥ 0 on $\mathbb{Q}$, hence on $\mathbb{R}$ by continuity, and its discriminant $a^2 - 4q$ is ≤ 0 . Either the roots are non-real and complex conjugate, or there is a double real root; in both cases $\beta = \bar{\alpha}$, and $|\alpha|^2 = \alpha\beta = q$. The zeros of ζ_E are the s with $q^{-s} \in \{1/\alpha, 1/\beta\}$ (the denominator does not vanish there), so $q^{-\operatorname{Re}(s)} = |q^{-s}| = q^{-1/2}$. The substitution $s \mapsto 1-s$ sends q^{-s} to $1/(q \cdot q^{-s})$, so $\zeta_E(1-s) = \zeta_E(s)$ is Problem 2.7(5). Compare [Sil09, Remark V.2.5].

(2) Taking logarithmic derivatives in Problem 2.7(4), written as $Z = (1-\alpha T)(1-\beta T)/((1-T)(1-qT))$, gives $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \alpha^n - \beta^n$, that is, $a_n = \alpha^n + \beta^n$ for $n \geq 1$; and $a_0 = 2 = \alpha^0 + \beta^0$. Then $|a_n| \leq |\alpha|^n + |\beta|^n = 2q^{n/2}$ by (1), and $n = 1$ is Hasse's bound.

(3) The series has coefficients bounded by $cq^{n/2}$, so it converges for $|T| < q^{-1/2}$ and defines a holomorphic function there. As formal power series, $\sum_n (\alpha^n + \beta^n)T^n = \alpha T/(1-\alpha T) + \beta T/(1-\beta T)$. If $\alpha \neq \beta$, the right side has poles at $1/\alpha$ and $1/\beta$, which do not cancel; if $\alpha = \beta$, it is $2\alpha T/(1-\alpha T)$, with a pole at $1/\alpha$. A holomorphic function on the disc $|T| < q^{-1/2}$ has no poles there, so $|1/\alpha|, |1/\beta| \geq q^{-1/2}$, that is, $|\alpha|, |\beta| \leq \sqrt{q}$. Since $|\alpha||\beta| = q$, both are equalities.

Problem 2.9 is [Sil09, Exercise 5.13].

Problem 2.9 (A recurrence).

- (1) Show that the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geq 1$.

- (2) Prove that

$$a_{n+2} = a_1 a_{n+1} - q a_n \quad \text{for all } n \geq 0.$$

(Recall the convention $a_0 = 2$.)

- (3) Here is the recurrence on $y^2 = x^3 + x + 1$ over $\mathbb{F}_{101}$:

```
q := 101; E := EllipticCurve([GF(q) | 1, 1]);           // y^2 = x^3 + x + 1
a := [TraceOfFrobenius(ChangeRing(E, GF(q^n))) : n in [1..6]];
a;                                                       // [ -3, -193, 882, 16847, -139623, -1282678 ]
b := [2, a[1]];
for n in [1..5] do Append(~b, a[1]*b[n+1] - q*b[n]); end for;
b[2..7] eq a;                                           // true
```

Here `TraceOfFrobenius(E)` is a . Compute a_2 by hand from $a_1 = -3$. Check Hasse's bound for a_6 over $\mathbb{F}_{101^6}$.

Solution.

(1) $\#E(\mathbb{F}_q)$ determines a , hence $Z(E/\mathbb{F}_q; T)$ by Problem 2.7(4), hence every $\#E(\mathbb{F}_{q^n})$. Explicitly, it determines α and β , and $a_n = \alpha^n + \beta^n$ by Problem 2.8(2).

(2) By Problem 2.8(2), $a_n = \alpha^n + \beta^n$ for $n \geq 0$. Since α is a root of $T^2 - aT + q$, $\alpha^{n+2} = a\alpha^{n+1} - q\alpha^n$, and likewise for β . Adding, $a_{n+2} = a a_{n+1} - q a_n$, with $a = a_1$. In particular every a_n is an integer, computed from a_1 alone.

(3) $a_2 = a_1^2 - 2q = 9 - 202 = -193$, as `Magma` says, and the recurrence reproduces all six traces. For a_6 : $2\sqrt{101^6} = 2 \cdot 101^3 = 2\,060\,602 \geq 1\,282\,678$.

Problem 2.10 (A zeta function by hand).

Let E be the elliptic curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_2$ (its discriminant is 1).

- (1) List the rational points of $E(\mathbb{F}_2)$, and find a .
- (2) Write down $Z(E/\mathbb{F}_2; T)$, and check the Riemann hypothesis for it directly.
- (3) Use Problem 2.9 to predict $\#E(\mathbb{F}_4)$ and $\#E(\mathbb{F}_8)$. Check the first by listing the rational points of $E(\mathbb{F}_4)$, where $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ with $\omega^2 = \omega + 1$.^a
- (4) Use Problem 1.2(4) to find the numbers N_1, N_2, N_3 of closed points of E of degree 1, 2 and 3. Deduce that every $\mathbb{F}_8$ -rational point of E is $\mathbb{F}_2$ -rational. Is $\#E(\mathbb{F}_8)$ close to the edge of the Hasse interval?
- (5) Check the Euler product of Problem 1.2(3) against (2) up to the term in T^2 .
- (6) Check, where `Places(E, d)` lists the closed points of degree d :

```
E := EllipticCurve([GF(2) | 1, 0, 0, 0, 1]);           // y^2 + xy = x^3 + 1
[#ChangeRing(E, GF(2^n)) : n in [1..8]];             // [ 4, 8, 4, 16, 44, 56, 116, 288 ]
[#Places(E, d) : d in [1..6]];                       // [ 4, 2, 0, 2, 8, 8 ]
ZetaFunction(E);                                     // (2*$.1^2 + $.1 + 1)/(2*$.1^2 - 3*$.1 + 1)
```

^a*Hint:* for $x \neq 0$, substitute $y = xz$.

Solution.

(1) For $x = 0$: $y^2 = 1$, so $y = 1$. For $x = 1$: $y^2 + y = 0$, so $y \in \{0, 1\}$. With O , $E(\mathbb{F}_2) = \{O, (0, 1), (1, 0), (1, 1)\}$, so $\#E(\mathbb{F}_2) = 4$ and $a = 2 + 1 - 4 = -1$.

(2) By Problem 2.7(4),

$$Z(E/\mathbb{F}_2; T) = \frac{1 + T + 2T^2}{(1 - T)(1 - 2T)}.$$

The roots of $T^2 + T + 2$ are $\alpha, \beta = (-1 \pm \sqrt{-7})/2$, with $|\alpha|^2 = (1 + 7)/4 = 2 = q$.

(3) With $a_0 = 2$, $a_1 = -1$ and $q = 2$: $a_2 = (-1)(-1) - 2 \cdot 2 = -3$ and $a_3 = (-1)(-3) - 2(-1) = 5$. So $\#E(\mathbb{F}_4) = 4 + 1 + 3 = 8$ and $\#E(\mathbb{F}_8) = 8 + 1 - 5 = 4$. To check the first: for $x = 0$, $y^2 = 1$ gives $y = 1$ only. For $x \neq 0$ put $y = xz$; dividing by x^2 , the equation becomes $z^2 + z = x + x^{-2}$. For $x = 1$ the right side is 0; for $x = \omega$ it is $\omega + \omega^{-2} = \omega + \omega = 0$ (as $\omega^3 = 1$); for $x = \omega^2$ it

is $\omega^2 + \omega^{-4} = \omega^2 + \omega^2 = 0$. So in each case $z \in \{0, 1\}$, giving two rational points each, and $\#E(\mathbb{F}_4) = 1 + 1 + 3 \cdot 2 = 8$.

(4) $N_1 = \#E(\mathbb{F}_2) = 4$, $N_2 = \frac{1}{2}(8 - 4) = 2$ and $N_3 = \frac{1}{3}(4 - 4) = 0$. By Problem 1.2(2), $E(\mathbb{F}_8)$ is the union of the orbits of size 1 and 3, and there are none of size 3; so $E(\mathbb{F}_8) = E(\mathbb{F}_2)$. The Hasse interval over $\mathbb{F}_8$ is $|a_3| \leq 2\sqrt{8} \approx 5.66$, and $a_3 = 5$ is as large as it can be: $E(\mathbb{F}_8)$ is as small as an elliptic curve over $\mathbb{F}_8$ allows.

(5) Modulo T^3 , the factors with $\deg P \geq 3$ are 1, so

$$\prod_P (1 - T^{\deg P})^{-1} \equiv (1 - T)^{-4} (1 - T^2)^{-2} \equiv (1 + 4T + 10T^2)(1 + 2T^2) \equiv 1 + 4T + 12T^2,$$

while $(1 + T + 2T^2)(1 + 3T + 7T^2) \equiv 1 + 4T + 12T^2$, using $1/((1 - T)(1 - 2T)) = \sum_n (2^{n+1} - 1)T^n$.

(6) The counts agree with $2^n + 1 - a_n$ for the recurrence of Problem 2.9, and the numbers of closed points with (4) and, beyond it, with Problem 1.2(4): for instance $N_4 = \frac{1}{4}(16 - 8) = 2$. Magma's zeta function is ours, with numerator and denominator written in increasing and decreasing order respectively: $2T^2 - 3T + 1 = (1 - T)(1 - 2T)$.

3. ISOGENOUS CURVES

Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$, with q -power Frobenius endomorphisms F_1 and F_2 and traces $a(E_1)$, $a(E_2)$. The Galois group $\text{Gal}_{\mathbb{F}_q}$ acts on each Tate module $T_\ell(E_i)$, and we write $\text{Hom}_{\text{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$ for the $\mathbb{Z}_\ell$ -linear maps that commute with this action.

Fact 3.1 (Tate's isogeny theorem). Let $\ell \neq p$ be prime. The natural map

$$\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \rightarrow \text{Hom}_{\text{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$$

is an isomorphism [Sil09, Theorem III.7.7(a)]. Recall also that $\text{Hom}(E_1, E_2)$ is a free $\mathbb{Z}$ -module of finite rank, being a subgroup of the free module of [Sil09, Corollary III.7.5].

Problem 3.2 is [Sil09, Exercise 5.4].

Problem 3.2 (Isogenous curves have the same number of rational points).

(1) Let $\varphi: E_1 \rightarrow E_2$ be a nonzero isogeny defined over $\mathbb{F}_q$. Show that $\varphi \circ F_1 = F_2 \circ \varphi$, and deduce that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

(2) Deduce that $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$.

For the converse, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $F_{1,\ell}$ and $F_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$ (Problem 2.7(3)). Fix a prime $\ell \neq p$ and put $V_i := T_\ell(E_i) \otimes \mathbb{Q}_\ell$.

(3) Show that a $\mathbb{Z}_\ell$ -linear map $T_\ell(E_1) \rightarrow T_\ell(E_2)$ commutes with the action of $\text{Gal}_{\mathbb{F}_q}$ if and only if it intertwines $F_{1,\ell}$ and $F_{2,\ell}$.

(4) Suppose $a^2 \neq 4q$. Show that $F_{1,\ell}$ is not a scalar, that V_1 has a basis $v, F_{1,\ell}v$, and that $V_1 \cong \mathbb{Q}_\ell[T]/(f)$ as $\mathbb{Q}_\ell[T]$ -modules, with T acting by $F_{1,\ell}$. Deduce that there is a nonzero $\mathbb{Z}_\ell$ -linear map $T_\ell(E_1) \rightarrow T_\ell(E_2)$ intertwining the two Frobenius maps.

(5) Suppose $a^2 = 4q$. Show that q is a square, that $a/2 \in \mathbb{Z}$, and that $F_i = [a/2]$ in $\text{End}(E_i)$ for $i = 1, 2$.^a Deduce that every $\mathbb{Z}_\ell$ -linear map $T_\ell(E_1) \rightarrow T_\ell(E_2)$ intertwines the two Frobenius maps.

(6) Use Fact 3.1 to conclude that E_1 and E_2 are isogenous over $\mathbb{F}_q$.

^aHint: compute $\deg(F_i - [a/2])$ with Fact 2.4.

Solution.

(1) The isogeny φ is given by rational functions with coefficients in $\mathbb{F}_q$, and raising to the q -th power fixes those coefficients: $\varphi(x^q, y^q) = \varphi(x, y)^q$ coordinatewise, that is, $\varphi \circ F_1 = F_2 \circ \varphi$. Hence $\varphi \circ (1 - F_1) = (1 - F_2) \circ \varphi$, and taking degrees, which are multiplicative, $\deg \varphi \cdot \deg(1 - F_1) = \deg(1 - F_2) \cdot \deg \varphi$. Since $\deg \varphi \neq 0$, Problem 2.7(2) gives $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

(2) The zeta function depends only on q and $a = q + 1 - \#E(\mathbb{F}_q)$, by Problem 2.7(4).

(3) For $Q \in E_i(\overline{\mathbb{F}}_q)$ we have $\sigma(Q) = F_i(Q)$, as in Problem 2.7(1); so σ acts on $T_\ell(E_i)$ by $F_{i,\ell}$. A map commuting with $\text{Gal}_{\mathbb{F}_q}$ therefore intertwines the Frobenius maps. Conversely, if λ commutes with σ , it commutes with every element of the subgroup $\langle \sigma \rangle$. The action of $\text{Gal}_{\mathbb{F}_q}$ on each $E_i[\ell^n](\overline{\mathbb{F}}_q)$ factors through a finite quotient $\text{Gal}(F/\mathbb{F}_q)$ for a finite extension F , which is generated by the image of σ ; so λ commutes with the whole group modulo ℓ^n for every n , and hence on T_ℓ .

(4) If $F_{1,\ell} = c$ were a scalar, its characteristic polynomial would be $(T - c)^2$, with discriminant $a^2 - 4q = 0$. So some $v \in V_1$ is not an eigenvector, and then v and $F_{1,\ell}v$ are linearly independent, hence a basis of the 2-dimensional space V_1 . The map $\mathbb{Q}_\ell[T] \rightarrow V_1$, $g \mapsto g(F_{1,\ell})v$, is therefore surjective; its kernel contains f by Cayley–Hamilton, and $\mathbb{Q}_\ell[T]/(f)$ has dimension 2, so $V_1 \cong \mathbb{Q}_\ell[T]/(f)$. The same holds for V_2 , so there is a $\mathbb{Q}_\ell[T]$ -isomorphism $\lambda: V_1 \rightarrow V_2$, that is, a linear isomorphism with $\lambda \circ F_{1,\ell} = F_{2,\ell} \circ \lambda$. The image $\lambda(T_\ell(E_1))$ is a finitely generated $\mathbb{Z}_\ell$ -submodule of V_2 , so $\ell^k \lambda$ maps $T_\ell(E_1)$ into $T_\ell(E_2)$ for k large, and $\ell^k \lambda$ still intertwines and is nonzero.

(5) Since $a^2 = 4q$ and $q = p^r$, the integer $a/2$ satisfies $(a/2)^2 = q$, so q is a square and $a/2 = \pm\sqrt{q} \in \mathbb{Z}$. Now $f(T) = (T - a/2)^2$, and by Fact 2.4

$$\deg(F_i - [a/2]) = \det(F_{i,\ell} - a/2) = f(a/2) = 0,$$

so $F_i = [a/2]$. Then $F_{i,\ell}$ is multiplication by $a/2$ on $T_\ell(E_i)$, and every $\mathbb{Z}_\ell$ -linear map commutes with multiplication by a scalar.

(6) In either case, by (3), there is a nonzero element of $\text{Hom}_{\text{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$, so by Fact 3.1 $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \neq 0$. As $\text{Hom}(E_1, E_2) \cong \mathbb{Z}^\rho$ is free, $\text{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \cong \mathbb{Z}_\ell^\rho$, so $\rho \geq 1$ and there is a nonzero isogeny $E_1 \rightarrow E_2$ defined over $\mathbb{F}_q$.

Remark 3.3. Together with Problem 2.7(4), Problem 3.2 says that for elliptic curves E_1, E_2 over $\mathbb{F}_q$ the following are equivalent: E_1 and E_2 are isogenous over $\mathbb{F}_q$; $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$; $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$; and E_1 and E_2 have the same number of closed points of each degree (Problem 1.2(4)).

Looking ahead. By Problem 2.8(1), $\alpha = \sqrt{q}e^{i\theta}$ for a unique $\theta \in [0, \pi]$, and $a = 2\sqrt{q}\cos\theta$. Fix an elliptic curve over $\mathbb{Q}$, reduce it modulo every prime p of good reduction, and let θ_p be the angle of the reduction over $\mathbb{F}_p$. How are the θ_p distributed in $[0, \pi]$ as p varies? That is the Sato–Tate conjecture. The product over the primes of good reduction of the factors $(1 - a_p p^{-s} + p^{1-2s})^{-1}$, the reciprocals of the numerators of the ζ -functions of the reductions, is, up to finitely many factors, the L -function of the curve, the subject of the Birch and Swinnerton-Dyer conjecture.

REFERENCES

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